

BSAN 6200: Text Mining & Social Media Analytics
Assignment 3: Topic Modeling Report

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Introduction

This report analyzes the Kaggle IMDb User Reviews dataset from Saad Azam using multiple text mining techniques to uncover themes and topics within the user reviews. This dataset consists of over 47,000 reviews across ten different movies, grouped by movies. After cleaning the data, and dropping corrupt reviews that could not be recovered, the topic modeling was applied at various k-values to find the highest coherence score. The code applied LDA and NMF topic modeling techniques to analyze sentiment patterns, create distinct groups and topics, and explore insights from an audience's perspective of various movies. The topic modeling techniques grouped the reviews into distinct topics and allowed for the creation of clear, direct, topic names based on the top words in each group and topic.

Data Collection & Preparation

The dataset was sourced from Kaggle and includes user reviews for ten movies: Morbius, Spider-Man: No Way Home, Pulp Fiction, The Avengers, Forrest Gump, Thor: Ragnarok, The Dark Knight, Joker, John Wick: Chapter 3 – Parabellum, and Avengers: Endgame. After combining all files, the dataset contained 47,063 reviews with associated metadata such as user ratings, review titles, and genres.

During data loading, approximately 2,197 reviews from The Dark Knight were excluded due to malformed rows caused by parsing issues. Since these records were unrecoverable, no imputation was performed. The final dataset remained sufficiently large for robust analysis.

Text preprocessing included lowercasing, removal of punctuation and stopwords, tokenization, and lemmatization using spaCy. Reviews with fewer than five tokens were removed, resulting in a cleaned dataset of approximately 42,000 reviews. Additionally, a review length feature was created, revealing an average length of approximately 144 words, with most reviews falling between 36 and 172 words.

Grouping Strategy

The dataset was divided into three meaningful groups based on movie genre and thematic similarity. The first group, labeled Group1_Superhero, includes Avengers: Endgame, Spider-Man: No Way Home, and The Avengers, with a total of 17,635 reviews. The second group, Group2_CrimeDrama, includes Joker, The Dark Knight, Pulp Fiction, and Forrest Gump, with 23,238 reviews. The third group, Group3_ActionOther, includes John Wick: Chapter 3 – Parabellum, Morbius, and Thor: Ragnarok, with 6,190 reviews.

Analysis of vocabulary size showed that the Crime/Drama group had the richest language usage with over 128,000 unique tokens, compared to approximately 95,000 for the Superhero group and 50,000 for the Action/Other group. This suggests that crime and drama films tend to generate more diverse and expressive reviews.

Topic Modeling Approach

Both LDA and NMF topic modeling techniques were applied separately to each group. For LDA, a model with 10 topics was used for each group, achieving coherence scores of approximately 0.51 for Superhero, 0.53 for Crime/Drama, and 0.55 for Action/Other. NMF models were also tuned across multiple values of k , with $k=10$ selected as the optimal number of topics based on coherence and reconstruction error. NMF produced higher coherence scores, reaching 0.6569 for Superhero, 0.6941 for Crime/Drama, and 0.7061 for Action/Other, indicating better interpretability.

Topic Interpretation by Group

Group 1: Superhero Movies

The Superhero group reveals topics centered around major Marvel Cinematic Universe elements. Prominent topics include “MCU Legacy,” “Thanos Snap,” “Infinity Holes,” and “Core Avengers,” reflecting strong discussion around overarching storylines such as Infinity War and Endgame. Topics such as “Spider-Men” and “No Way Home” highlight audience excitement around multiverse storytelling and character crossovers involving Spider-Man actors.

Other topics like “Action & CGI” and “Hero Storytelling” indicate that audiences frequently evaluate visual effects, large-scale battles, and narrative structure. Words such as “scene,” “fight,” “action,” and “battle” appear frequently, confirming that spectacle plays a major role in viewer engagement. At the same time, topics like “Villain Depth” show that audiences also critique character development, particularly for antagonists.

Overall, this group demonstrates that superhero movie discussions are heavily focused on interconnected storylines, major characters, and cinematic payoff, with strong emphasis on both spectacle and narrative continuity.

Group 2: Crime and Drama Films

The Crime/Drama group produces the most nuanced and diverse topics, reflecting deeper thematic engagement. Key topics include “Ledger’s Joker,” “Phoenix’s Joker,” and “Fleck’s Descent,” which emphasize character-driven discussions around performances by Heath Ledger

and Joaquin Phoenix. These topics include keywords such as “performance,” “oscar,” “actor,” and “mental,” highlighting the importance of acting quality and psychological depth.

Other topics such as “Nolan’s Realism” and “Gotham’s War” reflect appreciation for grounded storytelling and directorial vision, particularly in *The Dark Knight*. The presence of “Tarantino Cast” and “Pulp Fiction” topics indicates strong engagement with dialogue, nonlinear storytelling, and iconic characters.

Group 3: Action and Other Films

The Action/Other group is characterized by a strong emphasis on action sequences, choreography, and entertainment value. Topics such as “Wick’s World,” “Fight Craft,” and “Wick’s Stunts” reflect detailed discussions of combat, weapons, and choreography in *John Wick: Chapter 3 – Parabellum*. Keywords such as “fight,” “gun,” “knife,” and “choreography” appear frequently, indicating that viewers focus heavily on action realism and execution.

Other topics like “Ragnarok Cast” and “MCU Comedy” highlight the comedic tone and character interactions in *Thor: Ragnarok*, with frequent mentions of humor, actors, and director Taika Waititi. The “Asgard Lore” topic reflects interest in world-building and mythology.

The “Leto’s Morbius” topic reveals criticism and discussion surrounding *Morbius*, including references to characters and plot elements, often associated with weaker reception. Additionally, “Post-Credits” and “Visual Effects” topics show that audiences pay attention to technical elements and franchise-building strategies.

Overall, this group demonstrates that action-oriented films are evaluated primarily on entertainment value, action quality, and visual execution, with less emphasis on deep narrative complexity.

Additionally, the “Character Study” and “Mental Illness” themes suggest that audiences engage critically with deeper social and psychological issues. The “Forrest Gump” topic introduces themes of life events, history, and emotional storytelling, further emphasizing the diversity of this group.

Overall, this group highlights that crime and drama films generate more analytical and emotionally complex discussions, with strong focus on acting, character psychology, and thematic depth.

Key Findings

The results indicate that audience evaluation varies significantly by genre. Superhero films are judged based on their ability to deliver satisfying large-scale narratives and connect with broader cinematic universes. Crime and drama films are evaluated more critically in terms of acting performance, character depth, and thematic content. Action films are primarily assessed based on execution of action sequences and entertainment value. Additionally, NMF models consistently produced higher coherence scores than LDA, suggesting that NMF provided more interpretable and meaningful topics for this dataset.

Conclusion

In conclusion, this project demonstrates that text mining techniques can effectively uncover meaningful patterns in large-scale user review data. By grouping movies into three categories and applying topic modeling, the analysis reveals distinct ways in which audiences engage with different genres. Superhero films emphasize interconnected storytelling and spectacle, crime and drama films focus on character depth and realism, and action films prioritize choreography and entertainment. These findings provide valuable insights for filmmakers, producers, and analysts seeking to understand audience preferences and improve future productions.